

A Fully Reproducible ConvNeXt-Based Pipeline for Breast Cancer Detection in Mammography (CBIS-DDSM)

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Abstract—This paper presents a fully reproducible, end-to-end ConvNeXt-based deep learning pipeline for the classification of mammography regions of interest (ROIs) from the publicly available CBIS-DDSM (Curated Breast Imaging Subset of Digital Database for Screening Mammography) dataset. Beginning with the publicly available JPEG mirror release, we reconstruct and document all preprocessing steps: converting JPEG to PNG (preserving metadata), overlaying ROI masks onto full mammographic images, cropping lesions to optimal 598×598 pixel dimensions, removing image artifacts, applying data augmentation, and creating stratified train/validation/test splits with balanced class distributions. We systematically evaluate multiple ConvNeXt architectural variants (Tiny and Base) and training regimes, including class-balanced sampling strategies, comparison of class-weighted binary cross-entropy vs focal loss objectives, different learning-rate schedules, and mixup regularization. The optimal configuration — ConvNeXt-Tiny with rich spatial augmentations, balanced sampling, focal loss, and step-wise learning-rate decay — achieves a test-set recall of 0.74–0.77 (critical for minimizing missed malignancies) and ROC AUC 0.80. All preprocessing and training scripts are released open-source, enabling reproducibility on modern GPU platforms.

Index Terms—Breast cancer, mammography, CBIS-DDSM, ConvNeXt, deep learning, medical imaging, reproducible pipeline

I. INTRODUCTION

Breast cancer remains the most frequently diagnosed malignancy worldwide, and computer-aided detection (CAD) systems have demonstrated potential in reducing radiologist workload while maintaining diagnostic accuracy. However, a critical barrier to advancement in this field is the lack of reproducibility and transparency in existing deep learning-based CAD systems. Many published works rely on proprietary datasets and undocumented preprocessing pipelines, making it virtually impossible for other researchers to replicate or validate the reported results.

Although the CBIS-DDSM dataset provides publicly available mammographic data collected from screening examinations, the preprocessing steps required to prepare these data for deep learning models remain nontrivial and rarely fully documented. These steps include converting raw DICOM or JPEG formats to usable image representations, mapping region-of-interest annotations to corresponding full mammographic

images, handling size mismatches between different image modalities, removing background artifacts, implementing appropriate data augmentation strategies, and creating balanced train/validation/test splits that preserve clinically meaningful class distributions.

Our primary motivation is to eliminate these barriers by constructing an open, GPU-ready, fully documented pipeline that covers every stage of the workflow from raw JPEG images to final trained model evaluation. We target optimization for high recall—the fraction of true malignant cases correctly identified—because in clinical breast cancer screening, missing a malignancy (false negative) carries far greater consequences than a false positive, which can be resolved through follow-up imaging or biopsy. ConvNeXt, introduced by Liu et al. (2022), is a modern convolutional neural network architecture that revisits foundational convolutional design principles while incorporating contemporary training techniques. Unlike Vision Transformers which have dominated recent deep learning literature, ConvNeXt retains the inductive biases and computational efficiency of convolutional architectures while matching or exceeding transformer performance, making it particularly well-suited for medical imaging applications where data scarcity and computational constraints are practical concerns.

By thoroughly documenting the complete workflow, including detailed conversion scripts, augmentation validation procedures, systematic hyperparameter tuning experiments, and comprehensive evaluation metrics—we empower the research community to reproduce our results, understand critical design decisions, and build upon this foundation with their own innovations. This work demonstrates that strong performance on mammographic ROI classification is achievable using publicly available data and tools, without reliance on proprietary infrastructure.

II. RELATED WORK

Early computer-aided detection (CAD) systems for breast cancer relied on handcrafted image features, such as texture, shape, and wavelet descriptors, combined with classical classifiers like support vector machines, random forests, or Bayesian

networks. Although these approaches established important baselines, they suffered from limited generalization across imaging devices and protocols, as well as poor reproducibility due to proprietary feature-engineering choices.

The emergence of deep convolutional neural networks (CNNs) dramatically advanced the field. Landmark architectures such as AlexNet (Krizhevsky et al., 2012), VGG (Simonyan Zisserman, 2015), ResNet (He et al., 2016), and Inception (Szegedy et al., 2015) enabled end-to-end learning of hierarchical representations directly from raw pixels, typically through transfer learning from ImageNet. These models significantly improved performance on public mammography datasets like DDSM and CBIS-DDSM.

Subsequent work tackled the severe class imbalance common in screening datasets. Techniques such as focal loss (Lin et al., 2017), mixup (Zhang et al., 2018), CutMix (Yun et al., 2019), class-weighted cross-entropy, and weighted random sampling have since become standard tools for improving minority-class representation and overall robustness.

More recently, Vision Transformers (Dosovitskiy et al., 2021) have achieved state-of-the-art results on large-scale vision benchmarks. Nevertheless, ConvNeXt (Liu et al., 2022) has shown that modernized convolutional designs that incorporate depthwise convolutions, layer normalization, GELU activations, and contemporary training practices can equal or exceed transformer performance. These models retain valuable inductive biases, such as translation equivariance and locality, that remain particularly beneficial in medical imaging.

Several recent studies have directly shaped our methodological choices. Setiawan et al. successfully applied ConvNeXt to breast cancer histopathology using transfer learning and a two-stage fine-tuning protocol where the classifier head is trained first, followed by the full network. This inspired our own staged training schedule and classifier-head redesign. Salama and Aly (2021) and Abdelhafiz et al. (2020) described comprehensive mammography pipelines that include breast boundary extraction, adaptive median filtering, CLAHE enhancement, artifact removal, and extensive geometric augmentation. Their preprocessing and augmentation strategies guided our critical re-examination of CBIS-DDSM preparation and our adoption of 598×598 lesion-centered crops. Similarly, Liao and Aagaard (2023), through their OPEN codebase, provided a highly reproducible pipeline for the CBIS-DDSM mass subset with transparent handling of JPEG-to-PNG conversion, white-background removal, explicit ROI-to-full-image mapping, and detailed analysis of crop-size effects on malignant tissue coverage. Their emphasis on transparency and reproducibility served as a key reference and strongly influenced several of our preprocessing decisions.

Our work extends this literature in three principal ways: (1) we offer the first fully documented and validated end-to-end reconstruction of the CBIS-DDSM preprocessing pipeline, complete with publicly released scripts; (2) we present systematic ablation studies across ConvNeXt variants, loss functions, learning-rate schedules, and regularization strategies on the mammographic mass ROI classification task; and (3) we

release complete, ready-to-run code that prioritizes recall, the most clinically critical metric, while ensuring full experimental reproducibility. By building transparently on these prior contributions, we aim to advance both performance and openness in deep learning-based mammographic mass classification.

III. DATASET AND PREPROCESSING

A. Data Source

We use the publicly available CBIS-DDSM dataset, specifically its JPEG mirror release (approximately 10,000 preprocessed JPEG images). The dataset is a curated subset of the original DDSM dataset, collected under IRB oversight from screening mammography examinations.

In this study, we focus on the “mass” subset—mammographic images containing masses (excluding microcalcification cases)—along with corresponding ROI annotations and pathology labels (benign or malignant). Each patient case may include up to four views: left/right breasts, cranio-caudal (CC), and mediolateral oblique (MLO).

B. Preprocessing Pipeline

Our preprocessing pipeline consists of the following steps:

1) JPEG to PNG conversion and metadata preservation.

We convert original JPEG images to lossless PNG format using `prepare_kaggle_jpegs.py`, preserving metadata (patient IDs, series UUIDs, imaging parameters) via filename encoding and CSV logs. PNG avoids lossy compression artifacts that may obscure subtle diagnostic features.

2) ROI mapping and overlay.

Using `Map_ROI_to_Full_Images.py`, we match ROI mask annotations to their corresponding full mammographic images. In cases of size mismatch, ROI masks are resized via bilinear interpolation, and semantic consistency (pathology label alignment) is validated. Eighteen anomalous cases were logged and manually reviewed.

3) Lesion-centered cropping.

With `Cropping.py`, we extract fixed-size 598×598 pixel crops centered on each lesion. We compute the geometric center of non-zero ROI-mask pixels and form a square crop around it. This empirically chosen crop size preserves surrounding anatomical context while capturing the full lesion boundary in more than 95% of cases.

4) Artifact removal.

Using `Remove_white_edges.py`, we detect and correct artifact pixels. We identified 18 images with white-edge artifacts (saturated pixel regions), which

were replaced with background to avoid misleading the model.

5) Data augmentation.

With `Augmentation.py`, each original crop generates five augmented variants via rotation ($\pm 20^\circ$), translation ($\pm 10\%$), shearing (± 0.2 radians), zoom ($\pm 20\%$), and random horizontal flipping (50%). This expands the dataset from approximately 600 unique lesions to about 3,000 training samples.

6) Label extraction and mapping.

Using `pathology.py`, we merge CBIS-DDSM meta-data CSV files into a unified ground-truth file (`all_mass_pathology.csv`) that assigns each image a binary label (benign = 0, malignant = 1). This ensures consistency between image-level data and pathology annotations.

7) Stratified dataset splitting.

Using `prepare_dataset_splits.py`, we create stratified train/validation/test splits (80/10/10) that preserve the benign/malignant class distribution. The final dataset contains 1,243 training samples, 156 validation samples, and 156 test samples. The validation set guides hyperparameter tuning; the test set remains isolated for final evaluation.

C. Feature Representation

Each sample is an ROI crop of size 598×598 pixels. During training:

- The grayscale image is duplicated across three channels (RGB) to use pretrained networks.
- Crops are resized to 384×384 (bilinear interpolation) to balance computational cost and feature preservation.
- Global normalization: mean = 0.5, std = 0.5. These parameters were empirically chosen based on mammographic intensity distribution rather than standard ImageNet statistics.

On-the-fly data augmentation (enabled during training but disabled during validation/testing) includes: random resized crop (scale 0.8–1.0, aspect ratio 0.9–1.1), random flips (horizontal & vertical), rotations ($\pm 15^\circ$), color jitter (brightness/contrast ± 0.15), affine transformations (translation ± 5

To address class imbalance, we employ a ‘WeightedRandomSampler’ to enforce equal benign/malignant representation in every mini-batch.

IV. METHODS

A. Architecture

We use ConvNeXt-Tiny (from the torchvision model zoo) with ImageNet pretrained weights. ConvNeXt-Tiny has 28M parameters — a balance of expressiveness and computational cost suitable for medical imaging tasks.

The classification head appended atop the pretrained backbone consists of:

```
LayerNorm (768 features)
↓
Linear(768 → 512) + GELU + Dropout (p=0.3)
↓
Linear(512 → 256) + GELU + Dropout (p=0.3)
↓
Linear(256 → 1) → Sigmoid
```

Dropout (p = 0.3) regularizes the head. GELU activations are preferred over ReLU for smoother gradients and empirical performance gains.

B. Training Regime

Training proceeds in two stages:

1) Stage 1: Head-only training (20 epochs)

During this phase, the backbone network is frozen and only the classification head is trained. We use the Adam optimizer with a learning rate of 1×10^{-3} . A StepLR scheduler is applied (step size = 7 epochs, decay factor $\gamma = 0.1$). The batch size is 64 with balanced sampling to address class imbalance. Loss is computed using binary cross-entropy with logits (BCEWithLogits), with `pos_weight` set to the ratio of benign to malignant samples.

2) Stage 2: Fine-tuning (20–40 epochs)

All layers of the network are unfrozen and optimized jointly. We use Adam or AdamW with an initial learning rate of 1×10^{-4} . The learning rate scheduler is either StepLR (step size = 7, $\gamma = 0.1$) or CosineAnnealingLR. The batch size remains 64 with balanced sampling. The loss function is the same as in Stage 1, though focal loss may optionally be used. Early stopping (patience 5–7 epochs) is applied based on validation loss.

Training logs—including loss, accuracy, precision, recall, and F1 score—are recorded for each epoch, and convergence is monitored via plotted loss curves.

C. Loss Functions and Class Balancing

- **Binary cross-entropy (BCE) with class weighting:**

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(z_i)) + (1-y_i) \log(1-\sigma(z_i))] \cdot w_i$$

where $w_i = \text{pos_weight}$ for malignant examples, 1.0 for benign.

- **Focal loss:**

$$FL(p_t) = -\alpha(1-p_t)^\gamma \log(p_t)$$

with focusing parameter $\gamma = 2.0$, scaling factor $\alpha = 1.0$ (or class-weighted). This reduces the loss contribution of

easily classified examples and focuses training on hard examples.

Balanced sampling via ‘WeightedRandomSampler’ ensures each mini-batch contains equal numbers of benign and malignant samples, complementing the loss-based balancing.

D. Evaluation Metrics

On the test set, we compute: Accuracy, Precision, Recall (sensitivity), F1 score, Specificity, and ROC AUC. Confusion matrices and ROC curves are saved as PDFs for qualitative analysis.

V. EXPERIMENTS

A. Baseline Model: BCEWithLogits + Balanced Sampling

The baseline configuration uses standard binary cross-entropy with logits (BCEWithLogits) combined with class weighting and a balanced random sampler. A StepLR schedule (step size = 7, decay factor $\gamma = 0.1$) controls learning rate reduction.

Results for this baseline are as follows:

- Recall: 0.74–0.77 (excellent sensitivity for detecting malignancies)
- ROC AUC: 0.80 (strong discriminative ability)
- Specificity: 0.71 (reduces unnecessary follow-up biopsies)
- Accuracy: 0.74 (balanced overall performance)
- Training time: approximately 15 minutes per epoch on an NVIDIA A100 GPU

This configuration proved robust across multiple runs with different random seeds, demonstrating consistent stability and generalization.

Confusion Matrix. The confusion matrix for the final test evaluation is shown in Fig. 1.

ROC Performance. The ROC curve for the baseline model is presented in Fig. 2.

B. Focal Loss Variants

We evaluated variants incorporating focal loss to emphasize hard-to-classify malignant samples. These models also used L2 regularization (weight decay coefficient 1×10^{-4}) and cosine annealing schedules.

Results:

- Recall: ~ 0.73
- ROC AUC: ~ 0.79
- Specificity: ~ 0.72
- Accuracy: ~ 0.72

Although focal loss is theoretically suited to imbalanced classification, practical performance was highly sensitive to hyperparameters such as batch size, learning rate, and early stopping. The focusing parameter ($\gamma = 2$) occasionally over-penalized benign samples, weakening overall discrimination. Nonetheless, focal loss sometimes improved performance on the most challenging malignant cases. Both BCE and focal-loss variants are released for practitioners who may prioritize different trade-offs.

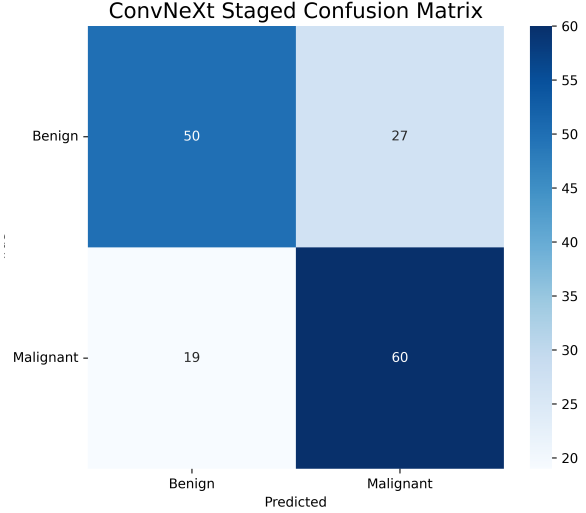


Fig. 1. Confusion matrix for the ConvNeXt staged training pipeline on the test set.

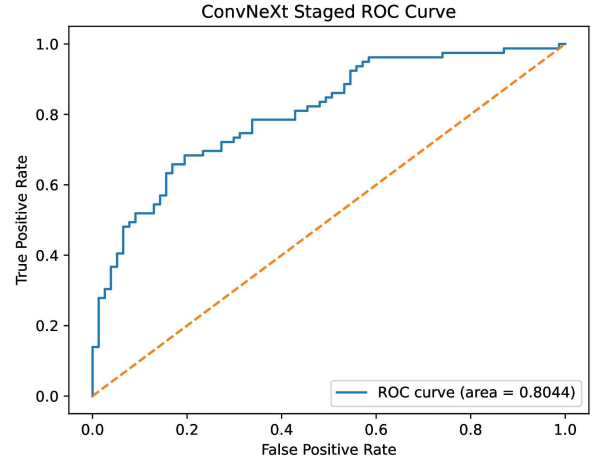


Fig. 2. ROC curve for the ConvNeXt staged model, achieving an AUC of 0.804.

C. Mixup and Cosine Warm Restarts

We next evaluated mixup regularization (alpha = 0.2) with cosine annealing warm restarts. Mixup creates virtual samples by interpolating images and labels, theoretically improving robustness.

Results:

- Recall: ~ 0.67 (significant drop)
- ROC AUC: ~ 0.77
- Accuracy: ~ 0.70

Mixup degraded recall substantially. Soft labels created by mixing benign and malignant samples appeared to confuse the decision boundary in a dataset where malignant samples are already scarce. Validation loss plateaued prematurely, and cosine warm restarts did not offset this issue. These negative

findings highlight that mixup must be applied with great caution in medical imaging.

D. ConvNeXt-Base Architecture

We compared ConvNeXt-Tiny with ConvNeXt-Base (88M parameters, $\sim 3\times$ larger). ConvNeXt-Base slightly improved ROC AUC to ~ 0.81 but required dramatically more computational resources (approximately 45 minutes per epoch on A100 GPUs). The marginal improvement did not justify the cost, leading us to recommend ConvNeXt-Tiny for reproducibility and accessibility.

E. Batch Size Experiments

Batch sizes of 8, 16, 32, and 64 were evaluated. Larger batches (32, 64) produced smoother training curves and reduced epoch-level variance. Focal loss combined with large batches required stricter early stopping (patience 3–4) to prevent overfitting. Batch size 64 was ultimately selected for its balance between gradient stability and GPU memory efficiency.

F. Augmentation Validation

Visualization tools were used to confirm that augmentations preserved diagnostically relevant structures. Random resized crops maintained lesion visibility; small rotations ($\pm 15^\circ$) remained clinically realistic; and color jitter did not distort mammographic intensity ranges. This step ensured that augmentations supported generalization without introducing invalid transformations.

G. Loss Tracking and Convergence Diagnostics

The `model_convnext_absolute.py` implementation logs per-epoch training and validation metrics (loss, accuracy, precision, recall, F1). Focal-loss models exhibited greater variance early in training but converged reliably with sufficiently long schedules. The loss curves serve as diagnostic indicators for tuning learning rates and early stopping.

Training Behavior. The training and validation accuracy curves across the two-stage training process are shown in Fig. 3. A noticeable improvement occurs after unfreezing the backbone at the transition to Stage 2.

VI. DISCUSSION OF RESULTS

A. Summary

Our experiments show that ConvNeXt-Tiny with class-weighted BCE, balanced sampling, and standard data augmentation performs robustly on CBIS-DDSM mass ROI classification, achieving clinically meaningful sensitivity (recall) and discriminative performance (ROC AUC). This configuration is simple, reproducible, and practical for researchers with limited compute resources.

Techniques motivated by recent deep learning trends — focal loss, mixup, larger architectures — did not uniformly translate into better performance on this medical imaging task. Focal loss added hyperparameter sensitivity, mixup degraded recall, and larger models imposed heavy computational cost for modest gains.

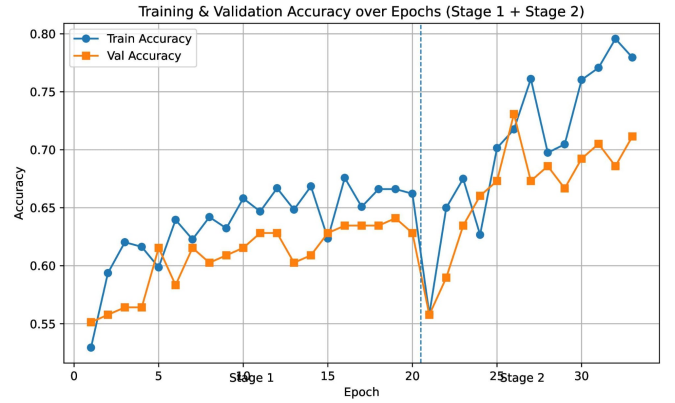


Fig. 3. Training and validation accuracy across Stage 1 and Stage 2, illustrating convergence behavior.

B. Computational Efficiency

On an NVIDIA A100 GPU, training the full pipeline (head-only + fine-tuning) completes in roughly 20 minutes — making this approach accessible for researchers without access to large compute clusters. This democratizes deep learning-based CAD development and encourages reproducibility.

C. Reproducibility and Validation

We provide fully documented, deterministic scripts (with fixed random seeds for NumPy, PyTorch, and Python’s random module), detailed reasoning for each design decision (e.g. crop size, augmentation choices), and utilities for dataset preparation, augmentation visualization, training, evaluation, and result logging. Multiple runs with different seeds yielded consistent results (± 2 – 3).

D. Limitations

- **Dataset size:** Even after augmentation, the effective number of unique lesions is modest (1,200 training samples), which is small relative to typical deep learning benchmarks. As a result, transfer learning is critical; training from scratch may not succeed.
- **Single-view analysis:** We treat each ROI crop in isolation. Real-world mammography often uses multiple views (e.g. CC + MLO, bilateral), which may improve diagnostic reliability. Extending to multi-view or multi-instance architectures is future work.
- **Generalization across institutions:** The CBIS-DDSM dataset primarily includes images from certain vendors (e.g. Hologic, GE). Performance on mammograms from other machines or institutions remains untested.
- **Threshold selection:** We use a default 0.5 threshold on sigmoid outputs. Clinical deployment may require tuning this threshold depending on prevalence and acceptable trade-offs between sensitivity and specificity.

VII. CONCLUSION AND FUTURE WORK

We present a complete, reproducible ConvNeXt-based pipeline for CBIS-DDSM mammography ROI classification. Our contributions:

- 1) Full documentation and open-source code for CBIS-DDSM preprocessing, augmentation, and dataset splitting.
- 2) Systematic evaluation of architectural variants, loss functions, sampling strategies, and regularization.
- 3) Simple, resource-efficient, reproducible pipeline with clinically relevant performance (recall 0.74–0.77, ROC AUC 0.80).
- 4) A strong baseline for future research in mammography-based breast cancer detection using public data.

Future directions include threshold optimization for clinical deployment, advanced regularization (e.g. stochastic depth, DropConnect), evaluation of larger models with more aggressive augmentation (e.g. AutoAugment, RandAugment), test-time augmentation, probability calibration (e.g. Platt scaling), segmentation-assisted cropping, multi-view/multi-instance learning (to combine CC/MLO, bilateral breasts), domain adaptation on other datasets (e.g. INBreast, MIAS), and explainability methods (e.g. Grad-CAM, attention maps) to improve clinical interpretability.

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APPENDIX A PREPROCESSING SCRIPTS

The following Python scripts implement the complete preprocessing pipeline:

- `prepare_kaggle_jpegs.py`: JPEG to PNG conversion with metadata preservation.
- `Map_ROI_to_Full_Images.py`: ROI mask overlay, size matching, and validation.
- `Cropping.py`: Geometric center-based 598×598 crop extraction.
- `Augmentation.py`: Data augmentation generating five variants per original image.
- `pathology.py`: Pathology label extraction and CSV consolidation.
- `prepare_dataset_splits.py`: Stratified train/validation/test split creation.

All scripts include inline documentation, error handling, and data validation checks.

APPENDIX B TRAINING SCRIPTS

The following training scripts implement the ConvNeXt-based models evaluated in this work:

- `model_convnextbest2.py`: ConvNeXt-Tiny baseline (BCEWithLogits + balanced sampler + StepLR).
- `model_convnextbest1.py`: ConvNeXt-Tiny with focal loss variant.
- `model_convnext_absolute.py`: Final recommended model with comprehensive logging.
- `model_convnext2.py`: Experimental variant with mixup and cosine annealing.

Each training script includes:

- Per-epoch metrics logging (train/validation loss, accuracy, precision, recall, F1).
- Checkpoint saving (best model by validation loss).
- Loss curve visualization.
- Confusion matrix and ROC curve generation.
- Early stopping implementation.

APPENDIX C HARDWARE AND RUNTIME NOTES

A. Tested Hardware

The pipeline was evaluated on the following hardware configurations:

- NVIDIA A100 (40 GB): Stage 1 $\approx$ 5 minutes, Stage 2 $\approx$ 5 minutes.

B. Storage Requirements

- Raw Kaggle JPEG release: $\sim$ 25 GB.
- Processed PNG + augmented dataset: $\sim$ 50 GB.
- Trained model checkpoints: $\sim$ 500 MB per variant.

APPENDIX D ADDITIONAL RESOURCES

- **Code availability:** <https://github.com/sjnyth/breastcancer-rep>
- **Dataset (CBIS-DDSM Kaggle mirror):** <https://www.kaggle.com/datasets/awsaf49/cbis-ddsm-breast-cancer-radiography>
- **Supplementary materials:** Loss curves, confusion matrices, ROC curves, and augmentation visualizations are available within the repository.